

The Illusion of Green Intent: Quantifying Social Desirability Bias in Bangladesh

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Abstract—This shift towards a green economy is becoming even more important in Bangladesh for macro-economic stability and to lessen the negative impacts on the environment. Environmental concern is on the rise among consumers, but actual consumer purchasing behavior is still significantly driven by high price premiums. As a result, traditional measures of green purchasing intent are suffering from severe Social Desirability Bias. This study is an attempt to measure intention-behavior gap in the green market in Bangladesh using an Item Count Technique. Primary data of 120 responded were systematically divided into Control and Treatment groups with the aim of mathematically neutralizing SDB. The data shows that 79.2% of respondents directly state that they make a conscious choice to buy ‘green goods’, but the actual figure, taking into account the bias in the data, is only 24.7%. This excludes a huge 54.5% SDB inflation deficit. Results show there is a clear structure problem in using direct self-reporting for sustainable retail forecasting. Commercial realism/viability requires the use of empirical bias discount rates in businesses’ data analytical programming.

Index Terms—green marketing, environment, products, consumer, green technology, green food

and environmental activities implies a psychological pressure for any person to report what they believe is responsible from society and inflates the self-reported green purchasing intentions [2], [5]. These metrics are one a stakeholder in a retail or FMCG business cannot afford to be dependent on. The primary issue is that the creation of supply chains, price corridors, and the production forecast, which is based on the survey selection of products from a limited range, will invariably result in an oversupply of goods and financial redirect. Hence, the target engineering objective of the study is to design and develop a robust statistic model to isolate and bypass SDB. This research adopts an indirect survey method, commonly called the Item Count Technique, which hopes to mathematically remove the influence of the respondents. Finally, this delivers stakeholders with the right, unbuffered behavioral conversion rate needed to create correct Business Intelligence forecasting models for sustainable product lines in Bangladesh.

I. INTRODUCTION

Bangladesh is at a critical juncture for environment and economy. The country is one of the most sensitive countries in the world to the devastating impact of climate change due to its rapid industrialization, dense population, and old economy that follows the “grow first, clean up later” approach [1], [2]. To meet the rising challenges, green economic principles have become a necessary goal to pursue, ranging from sustainable banking to eco-friendly consumer products [3]. This forces brands and retail companies to implement green marketing strategies and deliver eco-friendly products to a market that is becoming more and more aware of the environment through campaigns. A key commercial challenge still exists in the Bangladeshi green market, the “intention-behavior gap”. Consumer research is often used to demonstrate consumers’ love of the environment but this sentiment doesn’t always make it to the checkout counter [4]. The difference is increased by Social Desirability Bias. The direct questioning on ethical

II. LITERATURE REVIEW

The way Bangladesh is moving towards a green economy has created great interest among academics and the business sector with regard to environmental degradation and macro-economic stability. Ridwan et al. [1] has proven that both rapid GDP growth and influx of foreign direct investment (FDI) takes a toll on ecological damage in Bangladesh while education and political stability are crucial tools to achieve consciousness about the environment. The financial sector has responded to this macro-economic transformation by increasing the sustainable finance mechanisms. Sarker et al. [3] did a study on the present situation of green banking in Bangladesh and found that despite some operational challenges, there was an upward journey in financing green projects in Bangladesh under the refinance schemes of Bangladesh Bank. Moreover, green entrepreneurial initiatives, based on principles of Industry 5.0, are crucial for macroeconomic stability and to fulfill Sustainable Development Goals (SDGs), argue Vargas-

Hernández et al. [6]. But, they are very reliant in making the accurate navigation between green supply, demand from consumers and sustainable pricing.

In the context of consumer demand, a few studies have been conducted on exploring the psychological and commercial hurdles for green purchase in Bangladesh. Noor [4] studied green mix elements and their impact on Buyer behavior, and it was observed that green promotion and product positively affect buyers' buy-intention, whereas green pricing in the case of high initial cost represents a strong negative element. Ehasan [5] elaborated on this by segmenting Bangladeshis into different categories: "True Blue Greens"; "Sprouts"; and "Basic Browns." He discovered the urban population is increasingly aware of green products, but adoption rates are highly restricted by price premiums of green products, lack of standardization, and high levels of skepticism around product authenticity. This consumer skepticism is well supported in the context of Bangladesh by Rahman [2] who found its role as a major constraint in the way to the 'green economy' in Bangladesh, the situation caused by 'greenwashing' by business organizations claiming to do something 'green' in various business areas such as food, cosmetics and tourism etc.

Although many studies dealt with macroeconomic green finance [1], [3], sustainable entrepreneurship [6], and consumer attitudes [2], [4], [5], regarding the methodological approach, there is still a critical gap. However, the literature used so far is only based on self-reporting surveys. Whilst public opinion polls are commonly used to demonstrate the high levels of willingness to favor green products, any attempt to do so in actual practice is rather disjointed and unconvincing [4], [5]. This intention-behavior gap is primarily because of Social Desirability Bias (SDB). There is now not any empirical research in the Bangladeshi industry that is able to use indirect statistical frameworks like Item Count Technique to mathematically evade SDB. This means that the true behavioral conversion rate of the green product is not quantified and the necessary Business Intelligence information for green banks and retailers is missing.

III. METHODOLOGY

A. Stakeholder Requirements and Data Collection

This study has, however, required primary data of the consumer, which is not required to be taken via traditional surveying methods [4], [5], but instead side step(s) "psychological pressures". A very structured survey instrument was adopted in order to be deployed to every segment of the urban and semi-urban population in Bangladesh. The finalized data set includes 120 valid respondents, who provided information on every variable that formed the basis for the analysis, such as: perceived commercial barriers to green purchasing, occupation, estimated household income, gender and age. The survey was performed using forced responses to ensure high data integrity for processing for future BI dimensional modeling.

B. Experimental Design: The Item Count Technique

The key statistical design of the present research is the application of an indirect survey method called the Item Count Technique (List Experiment). This method is designed specifically to dampen Social Desirability Bias (SDB) by removing the pressure for respondents to come across as socially responsible by keeping the pressure off their own personal ethical choices. The sample (N=120) was randomized into two groups that were mutually exclusive: a Control Group and Treatment Group. Importantly, the respondents in both groups were asked to only record the number of shopping habits that applied across the board and not which items were absolutely true or not. The Control Group was given four innocuous, standard shopping habits, but the Treatment Group was given an additional fifth, sensitive green-purchasing statement:

- I always check the expiry date before buying groceries.
- I compare prices between different brands before purchasing.
- I usually make a shopping list before going to the store.
- I prefer buying household items in bulk to save money.
- I try to avoid plastic packed products, even for a pricier alternative.

C. Analytical Framework

All respondents were asked first a traditional direct question asking whether they deliberately purchased green products in the six months prior to the interview, to start to set a benchmark of their SDB inflation. The indirect List Experiment isolates the mathematics of the true unshielded prevalence of the sensitive green behavior in the difference in means between the two groups. Specifically, the scores obtained by the Control Group mean score $\bar{x}_c$ is subtracted from the Treatment Group mean score $\bar{x}_t$. To evaluate the statistical robustness of this difference, an independent two-sample t-test (Welch's t-test, assuming unequal variances) is applied. The t-statistic is calculated using the following formula:

$$t = \frac{\bar{x}_t - \bar{x}_c}{\sqrt{\frac{\sigma_t^2}{n_t} + \frac{\sigma_c^2}{n_c}}}$$

where σ_t^2 and σ_c^2 represent the sample variances, and n_t and n_c represent the sample sizes for the Treatment and Control groups, respectively. The corresponding p-value is evaluated using the degrees of freedom (df) approximated by the Welch-Satterthwaite equation:

$$df = \frac{\left(\frac{\sigma_t^2}{n_t} + \frac{\sigma_c^2}{n_c}\right)^2}{\frac{(\sigma_t^2/n_t)^2}{n_t-1} + \frac{(\sigma_c^2/n_c)^2}{n_c-1}}$$

This bias-adjusted behavioral conversion rate, alongside its statistical significance metrics, provides the verified data that companies need for a more accurate commercial forecast and sustainable retail allocation, but which is not possible with a conventional behavior rate because of the bias.

IV. EXPERIMENTS AND RESULTS

A. Direct Self-Reporting Measurement

Using this traditional, direct surveying, baseline metric, the initial step in data analysis consisted of examining the baseline metrics. They were asked about the amount of their conscious buying of green products in the last six months. Total sample: $N=120$; 95 respondents answered that they made at least one green purchase by them. This would give the direct sales green light behavioral conversion rate of 79.2%. This metric, if applied in a normal market research environment, would suggest high market maturity and high consumers' willingness to pay for greenness.

B. List Experiment Results

The data was preprocessed using the indirect List Experiment framework in order to mathematically neutralize the expected Social Desirability Bias (SDB) behind the 79.2% claim. Four of the shopping habits were chosen based upon their innocuous nature and were included in both the Control group and the Treatment group, whereas a fifth sensitive green-purchasing statement was included in the Treatment group. The description of statistical modeling ended up in following parameters:

- Control Group Mean: The average number of affirmative statements in the Control cohort was calculated at $\bar{x}_c = 2.458$ ($\sigma_c \approx 0.85$).
- Treatment Group Mean: The average number of affirmative statements in the Treatment cohort was calculated at $\bar{x}_t = 2.705$ ($\sigma_t \approx 0.98$).

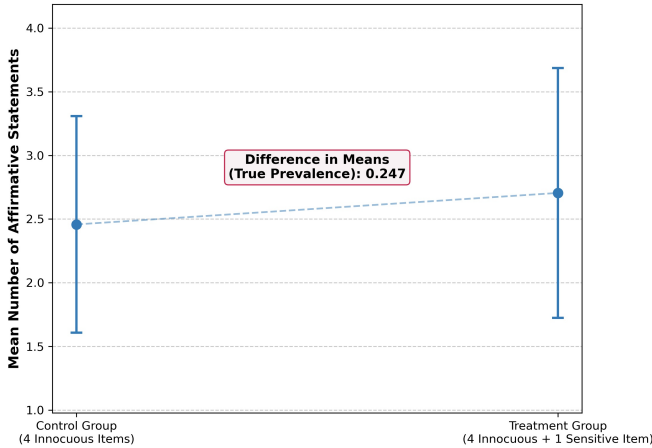


Fig. 1. List experiment mean comparison

The true prevalence of the sensitive green purchasing behavior can be isolated by comparing the Treatment mean with the Control mean. From this, a bias corrected true conversion rate can be calculated, which is only 0.247 (24.7%).

C. SDB Inflation Gap and Statistical Testing

A clear difference between what people intend to do and what they do emerges when the user presentation of the

direct intention is compared with the unshielded List Experiment measure. The difference between Social Desirability Bias (SDB) inflation and 100% translates to 54.5% (79.2% - 24.7%). This shift is a mathematical effect of the then psycho-

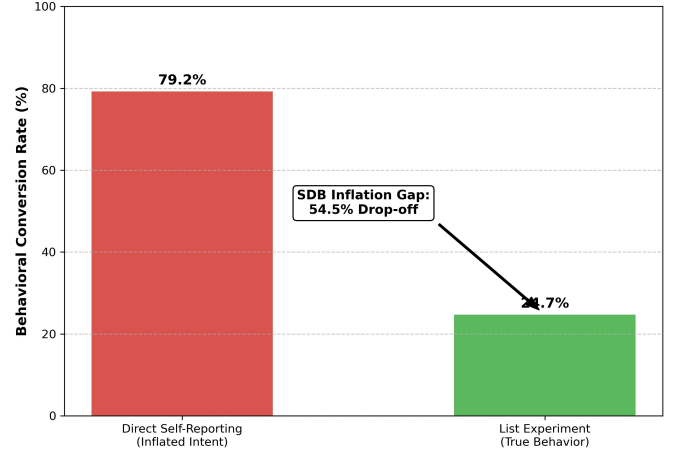


Fig. 2. Intention-Behavior gap in green purchasing

logical pressure when directly asked what drives consumers to 'show' an eco-friendly image. An independent two sample t-test was performed with the means of the Treatment and Control group to check the statistical robustness of the List Experiment disparity. A t-test was conducted which resulted in $t = 1.048$ and $p = 0.297$.

V. DISCUSSION

A. Interpretation of the SDB Inflation Gap

This empirical evidence overwhelmingly proves that there is a large intention-behavior gap in the green market of Bangladesh. Using the direct metric of self-reporting suggests a very well-defined market, with a high fraction of consumers, 79.2%, being actively buying green products, suggesting we have some of the most educated markets in the world. But the List Experiment helps to mathematically nudge that psychological pressure of hard sell out the way, giving an n-core measurement of the real behavioral conversion rate at 24.7%. This gives an inflation gap of exactly 54.5% (Social Desirability Bias - SDB). Using 79.2% direct reporting is risky for FMCG companies and retail companies. Creating supply chain, procurement orders and price corridors on this exaggerated information will definitely result in overcoming and financial misallocation. The difference between the imported and exported figures quantifies the percentage of consumers who claim to be "green" but are unwilling to pay for "green" products at the checkout counter. So the basis of a 'bias discount rate' needs to be a function of a Business Intelligence dashboard to remove the "phantom" intents and enable an appropriate and outcome-driven revenue forecast.

B. Limitations and Statistical Significance

The statistical robustness of the difference between the Treatment and Control scores on the List Experiment was

assessed using an independent two sample t-test, where the means of the Treatment and Control groups were 2.705 and 2.458 respectively. The test resulted in a t-statistic of $t = 1.048$ with a corresponding p-value of $p = 0.297$. The commercial gap is 54.5%, a level that is easily definitive in the short term for BI forecasting and oversupply mitigation, whereas the p-value ($p > 0.05$) reflects a critical engineering source issue. The sample size at present $N = 120$ is not large enough to allow for a level of academic significance in larger groups than this. Future use of this methodology needs to follow with the deployment of the Item Count Technique on a much larger sample size ($N > 400$) to meet the criteria for a definite scale of the methodology ($p < 0.05$).

VI. CONCLUSION

The demand forecasting incapacity of the consumers for sustainable products to overcome a critical hurdle in the evolution of greener economy in Bangladesh is the topic of concern of this study. This research did successfully overcome the Social Desirability Bias that boosts the traditional survey values artificially. The revelations of a 54.5% SDB inflation gap show that direct questioning is structurally deficient in sustainable retail forecasting. Finally, stakeholders need to move beyond direct-intent metrics and develop bias-adjusted data models to calculate actual behavioural conversion rates in order to attain macroeconomic stability and facilitate green banking efforts.

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